Fiscal Year:	FY 2011	**Task Last Updated:**	FY 04/13/2011
PI Name:	Limoli, Charles Ph.D.		
Project Title:	Oxidative stress as a mechanism for altering acute and chronic functional changes in the CNS exposed to low dose, low dose rates, and mixed fields of protons and HZE nuclei		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation Biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	climoli@uci.edu	**Fax:**	FY (949) 824-3566
PI Organization Type:	UNIVERSITY	**Phone:**	(949) 824-3053
Organization Name:	University of California		
PI Address 1:	Dept. of Radiation Oncology		
PI Address 2:	Medical Sciences I, B149		
PI Web Page:			
City:	Irvine	**State:**	CA
Zip Code:	92697-2695	**Congressional District:**	45
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Space Radiobiology NNJ08ZSA001N
Start Date:	07/01/2009	**End Date:**	06/30/2012
No. of Post Docs:	2	**No. of PhD Degrees:**	2
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
Contact Email:	Yuri.V.Griko@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Huang, Ting-Ting (VA Palo Alto Health Care System) Elmore, Eugene (University of California Irvine)		
Grant/Contract No.:	NNX09AK25G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The space radiation environment poses unique hazards that include solar particle events and galactic cosmic rays. These types of radiations have been shown to cause acute and long-lasting chemical changes referred to as oxidative stress that can alter the normal physiology of the central nervous system (CNS). The chemical species produced can cause a wide range of effects in cells within the CNS that largely depend on the types of pathways they activate and the types of damage they produce. Proper functioning of the CNS depends on many factors, but is known to require subpopulations of neural stem and precursor cells that participate in the repair of CNS injury resulting from irradiation. Our proposal is focused on analyzing the effects of space radiation on these critical populations of multipotent neural stem and precursor cells. We have developed the technology to isolate and grow these cells in culture and will use a variety of techniques to analyze how cells exposed to space radiation change. We will monitor the time course of oxidative stress in these cells and how that depends on the dose, the dose rate and type of radiation applied. We will determine how radiation-induced oxidative stress impacts cellular proliferation, signaling between cells, cell fate and survival. Collectively our studies using both rodent and human neural stem and precursor cells will help NASA understand the acute and longer-lasting sequelae associated with the exposure of the CNS to the space radiation environment. These studies will facilitate efforts to more appropriately establish estimates of risk for radiation exposure to the CNS in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Results from these studies will be critical in enhancing our knowledge base of the risks associated with exposure of the brain to the radiation fields in space.
Task Progress:	Scope of Work - General We have initiated a number of the experiments proposed in our currently funded NASA grant. Our overarching goal is aimed at determining if/how low dose and low dose rate proton and HZE charged particle irradiation leads to oxidative stress in neural precursor cells, and whether oxidative stress constitutes a biochemical mechanism responsible for disrupting cellular physiology and promoting cognitive dysfunction. Our past data has demonstrated that the radiation response of multipotent neural precursor cells to x-rays, protons, and HZE nuclei involves an acute and persistent oxidative stress that can last for months. The importance of neural precursor cells to the repair and remodeling of the irradiated CNS is underscored by our data showing that the radiation-induced depletion of these cells is linked to the inhibition of neurogenesis and the onset of cognitive impairment. We believe oxidative injury is critical in this context, and plays a contributory if not causal role in determining neural precursor cell function, survival and fate. We have shown that oxidative stress significantly alters proliferation, radiosensitivity, mitochondrial function, and differentiation, and the studies proposed in this grant will extend these studies to space relevant irradiation paradigms. Significant new findings and accomplishments under the past reporting period are as follows: 1) We have published a comprehensive study defining the radiobiological response of human neural stem cells to low LET radiation modalities. Acharya, M., Lan, M., Kan, V., Patel, N., Giedzinski, E., Tseng, B.P. and Limoli, C.L. Consequences of ionizing radiation-induced damage in human neural stem cells. Free Radic. Biol. Med. 49:1846-1855 (2010). 2) We have had 2 successful BNL runs and 4 successful Loma Linda runs with these cells and are accumulating data regarding their response to charged particle and proton irradiation (see below). 3) We have generated neural stem cells from mice deficient for the extracellular isoform of superoxide dismutase (EC-SOD). Studies have now been initiated to characterize how cells lacking this key antioxidant enzyme respond to low and high LET radiation (see below). 4) We have successfully optimized the assessment of metabolic activity after low dose low LET exposure using non-invasive 2-photon spectroscopy. This manuscript is currently under submission: Krasieva, T.B., Giedzinski, E., Tran, K., Lan, M.L., Limoli, C.L. and Tromberg, B.J. Probing the impact of gamma-irradiation on the metabolic state of neural stem and precursor cells using dual-wavelength intrinsic signal two-photon excited fluorescence. Submitted 3/21/2011 to J. Innovative Optical Health Sciences. 5) Significant new work with combinations of protons and gamma ray exposures at Loma Linda have uncovered evidence for radiation-induced adaptive responses in human and rodent neural stem cells that are both dose-dependent and dependent on the interval between priming and Challenge doses (see below). 6) We have expanded our ultra low fluence studies (= 1 particle traversal/cell) using silicon (300, 600 1000 MeV/amu), oxygen (300, 1000 MeV/amu) and iron (300, 600 1000 MeV/amu) ions. These studies have now indicated a trend suggesting that acute increases in radiation-induced oxidative stress depend on particle energies while more chronic increases in radiation-induced oxidative stress follow the particle LET. 7) Radioadaptive studies are ongoing and show that prior exposure alters radiation-induced oxidative stress. This stress has been found to in general, improve survival, enhance DSB repair, and elevate the intracellular antioxidant capacity of neural stem and precursor cells. 8) Additional work including new ions (Si, O) and energies (300, 600, 1000 MeV) has added to our prior findings demonstrating the persistence (weeks) of oxidative stress after low dose charged particle irradiation. Studies planned for the upcoming year will take advantage of newly optimized assays and cell systems and will continue to utilize mixed fields of protons and iron ions to more faithfully simulate the complexities of the space radiation environment. As such, the research proposed below has relevance to the estimation of early and late CNS risks involved in space exploration. As in the past, we will continue to focus our work on low doses and fluences of protons and HZE nuclei for more accurate gauges of how realistic exposure scenarios elicit changes in the redox microenvironment of the CNS that impact stem cell function, fate, and loss.
Bibliography Type:	Description: (Last Updated: 01/14/2026)

Articles in Peer-reviewed Journals	Acharya MM, Lan ML, Kan VH, Patel NH, Giedzinski E, Tseng BP, Limoli CL. "Consequences of ionizing radiation-induced damage in human neural stem cells." Free Radical Biology and Medicine. 2010 Dec 15;49(12):1846-55. Epub 2010 Sep 6. PMID: 20826207 , Dec-2010